

ACADEMIC NLP PROJECT

Ro-Invekt

A Multi-Stage Hybrid Pipeline for Robust Romanian Hate Speech Detection

Authors: Jalba Cosmin · Tudor Apostol · Albert Cohen · Agusoaei Alexandru

Research Problem

Linguistic Complexity

Romanian hate speech uses slang, inflections, and creative profanity that bypass simple filters.

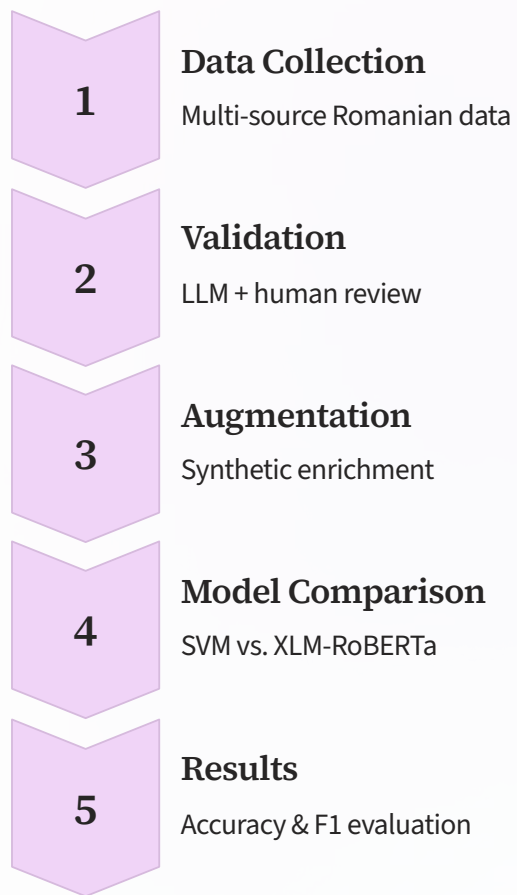
Data Scarcity

Balanced Romanian toxicity datasets are scarce, limiting model training quality.

Keyword Filters Fall Short

Context-aware models are needed — keyword-only approaches miss nuanced hate speech.

Project Goal



Build a robust Romanian toxicity classification pipeline combining multi-source data, rigorous validation, and augmentation — benchmarking Linear SVM against XLM-RoBERTa.

Data Sources

Platforms

Reddit, Twitter, Facebook Live, and Romanian news comments.

Corpora

- Hate-Speech-RO
- OLID-ROFF
- News-RO-Offense
- RO-FB-Offense

Corpus Size

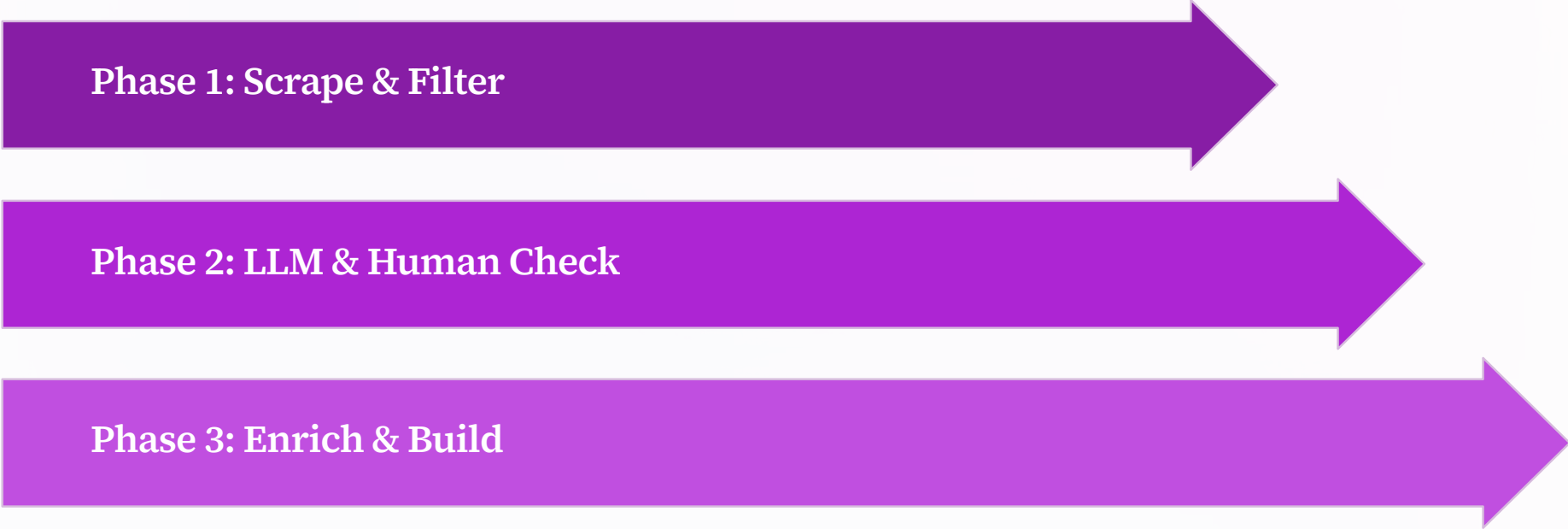
~10,000–15,000 labeled rows across all sources.

Hate Speech Taxonomy

Classification Levels

- **Neutral** — Non-toxic content
- **General Aggression & Insults** — Broad hostility
- **Hate Speech & Discriminatory** — Targeted group harm
- **Misogyny, Ableism, Family/Religious Invective** — Identity-based attacks
- **Politics, Sports & Rivalry** — Contextual aggression

Three-Phase Pipeline



Phase 1: Scrape & Filter

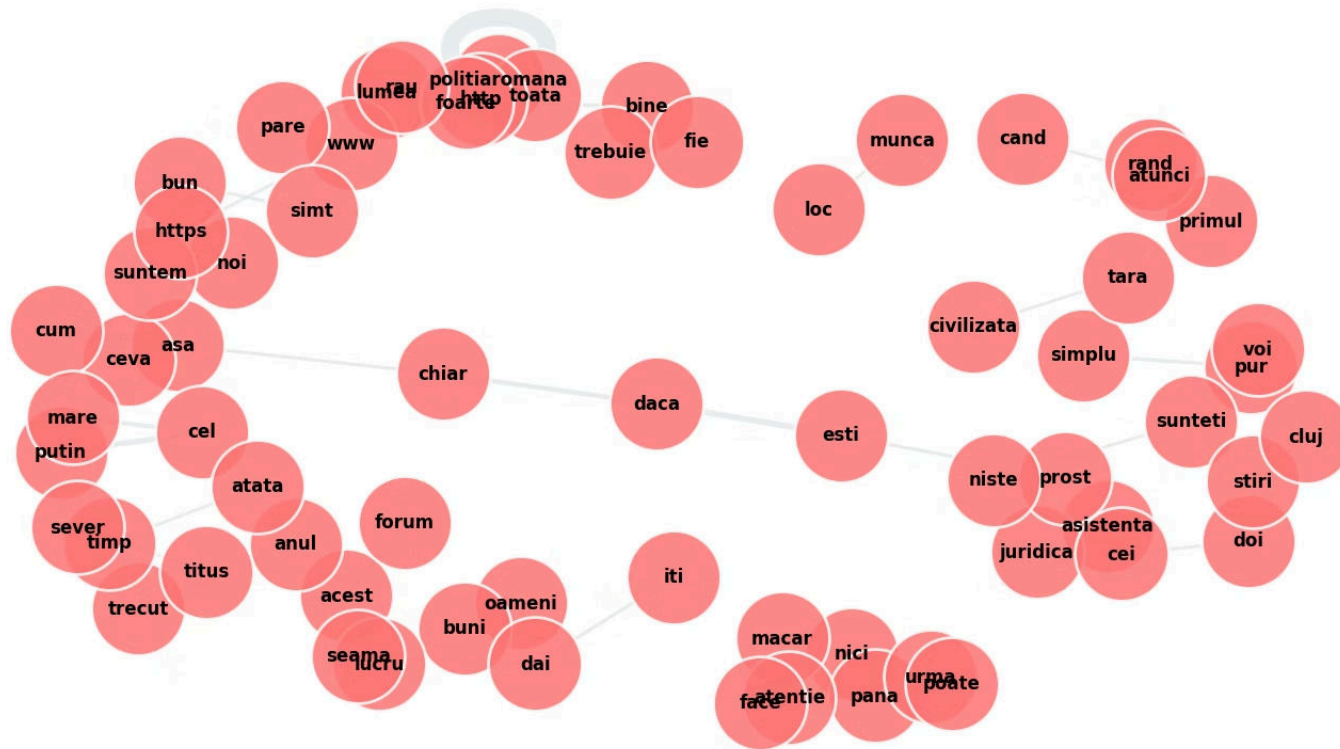
Phase 2: LLM & Human Check

Phase 3: Enrich & Build

Each phase progressively refines raw data into a high-quality, validated, and augmented Romanian hate speech corpus ready for model training.

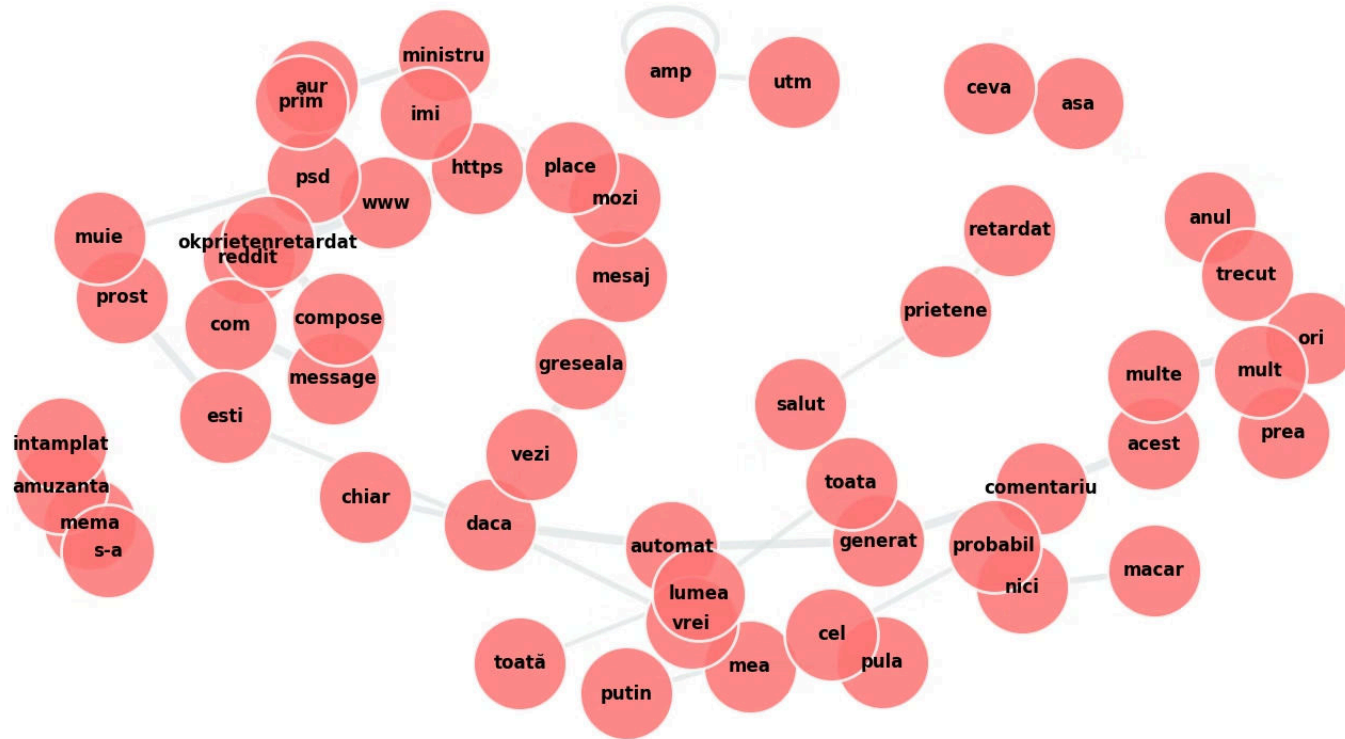
News-RO Analysis

Network visualization (Louvain clustering) - top 10 clusters



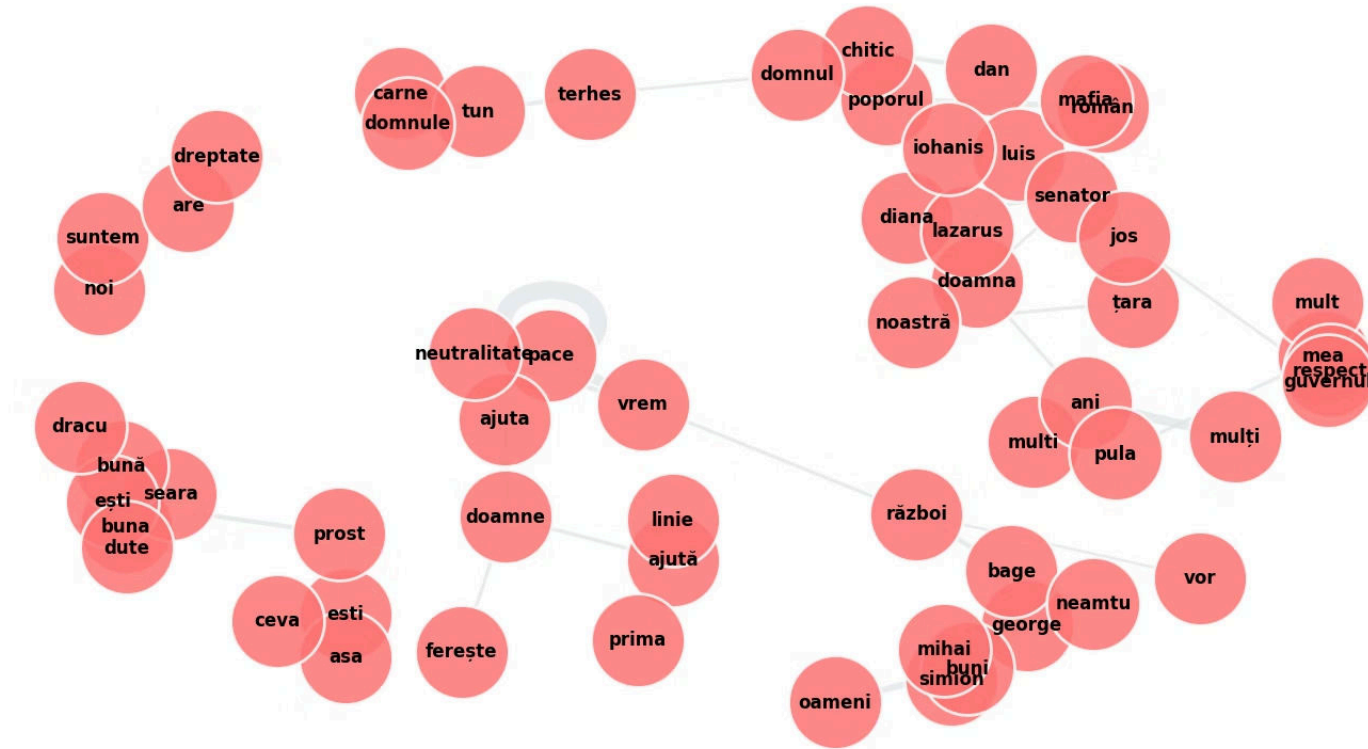
Reddit Analysis

Network visualization (24/04/2019, 10:10:10)



RO-Facebook Analysis

Mapa conexiunilor (edges) dintre cuvinte (nodes)

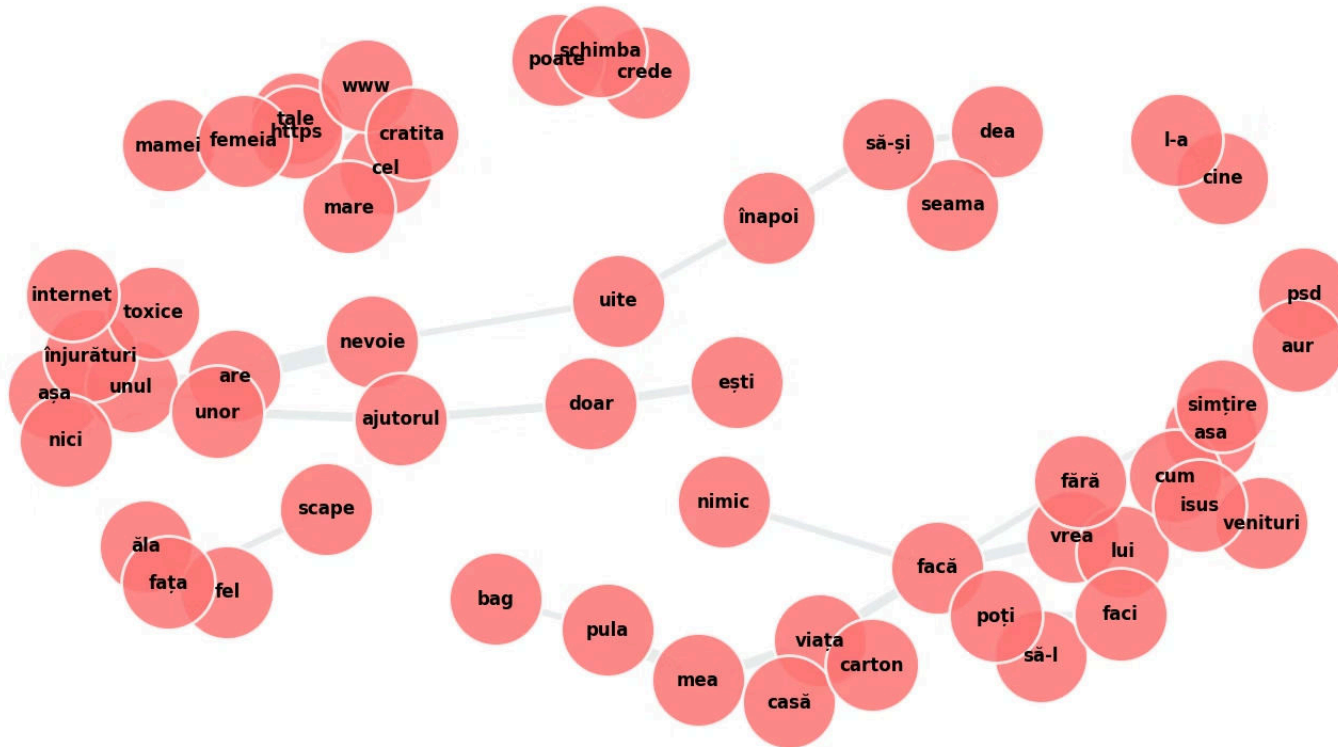


ROFF Analysis



V8 Balanced Dataset

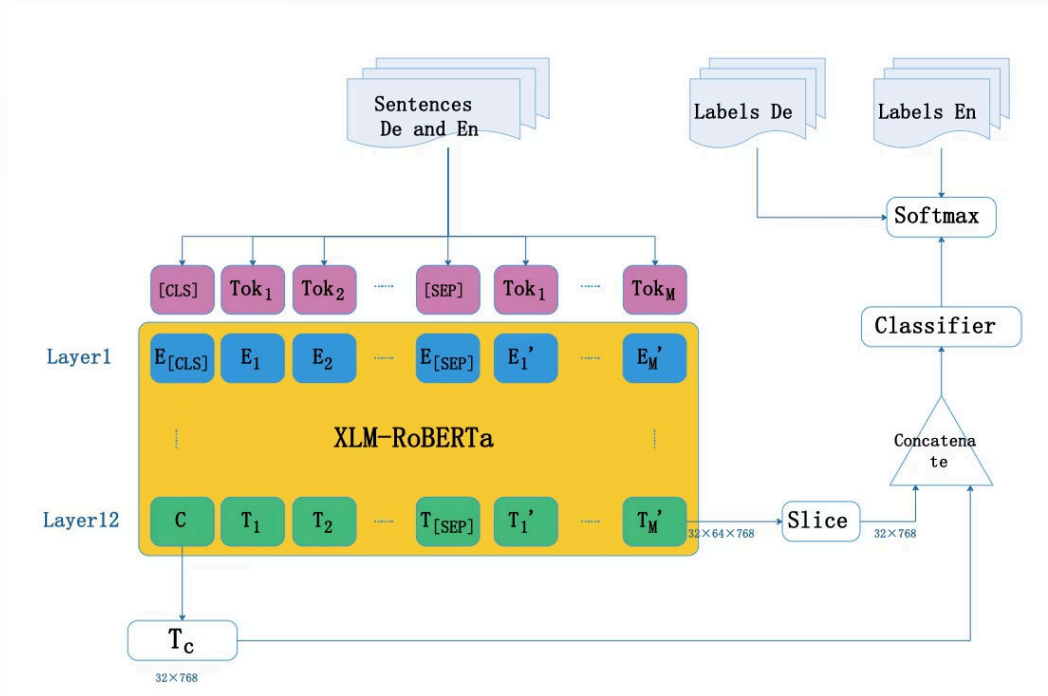
Marcel Cernăuțanu, Vlad Căstănel, Vlad Căstănel, Vlad Căstănel



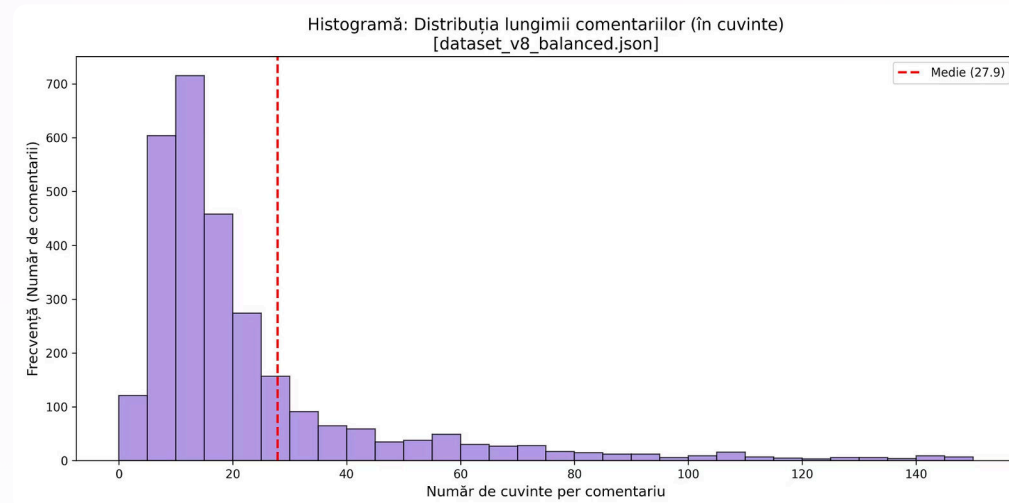
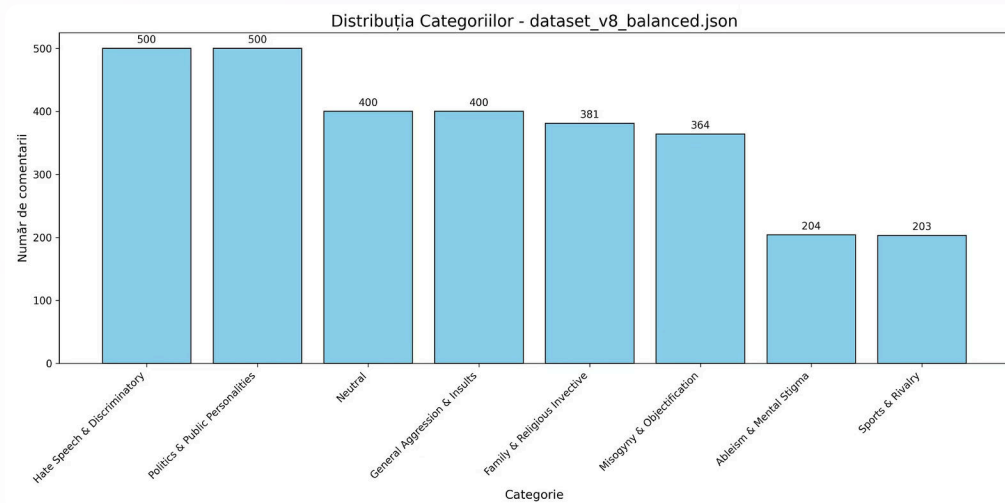
Models & Results

Dataset	Model	Accuracy	Macro F1
High-Synthetic 80%	SVM	42.1%	0.38
High-Synthetic 80%	XLM-RoBERTa	45.0%	0.40
Reddit Filtered	SVM	70.0%	0.64
Reddit Filtered	XLM-RoBERTa	76.3%	0.71
Augmented 30%	XLM-RoBERTa	72.5%	0.69
Augmented 60%	XLM-RoBERTa	74.4%	0.73

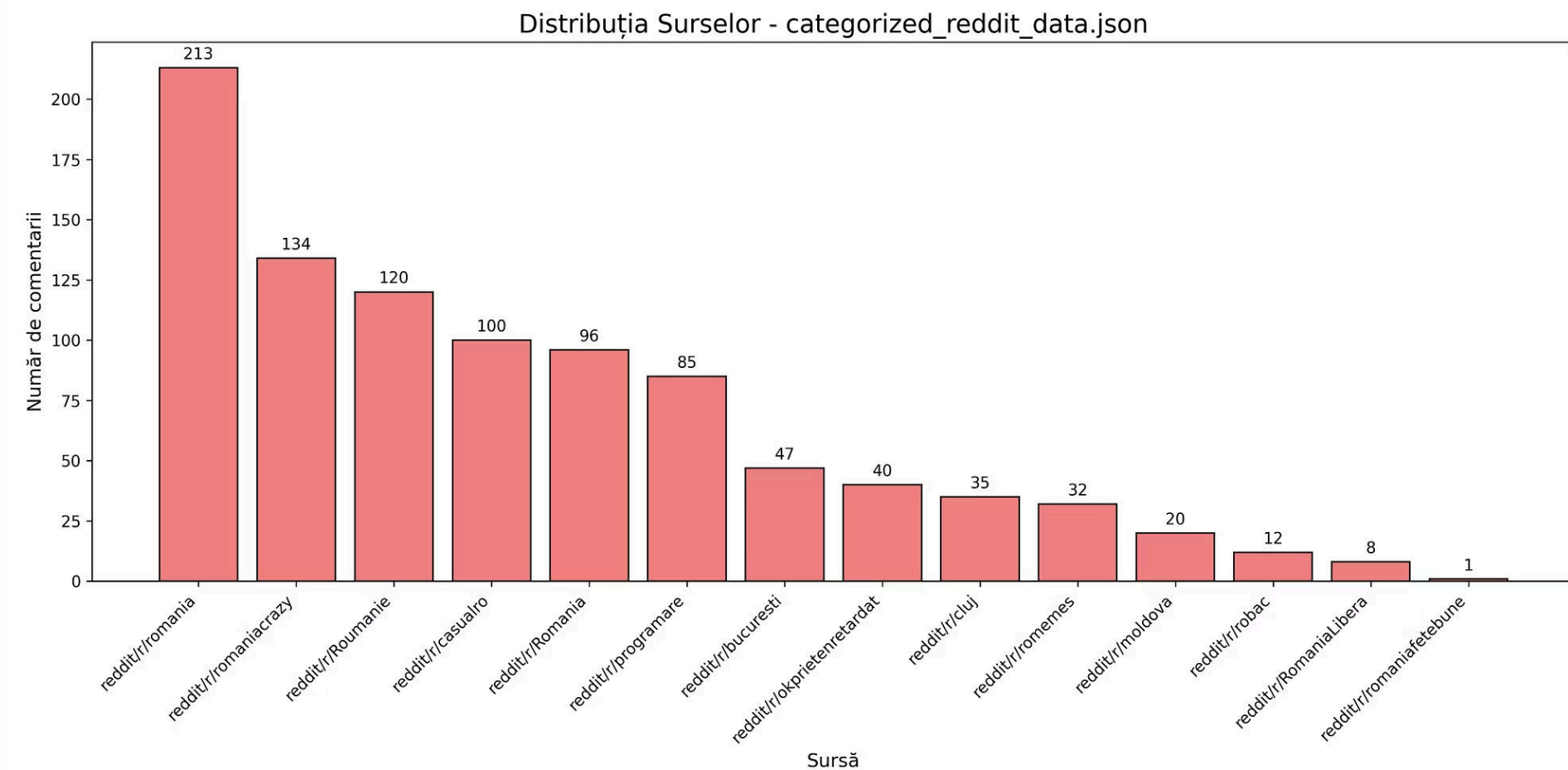
XLM-RoBERTa



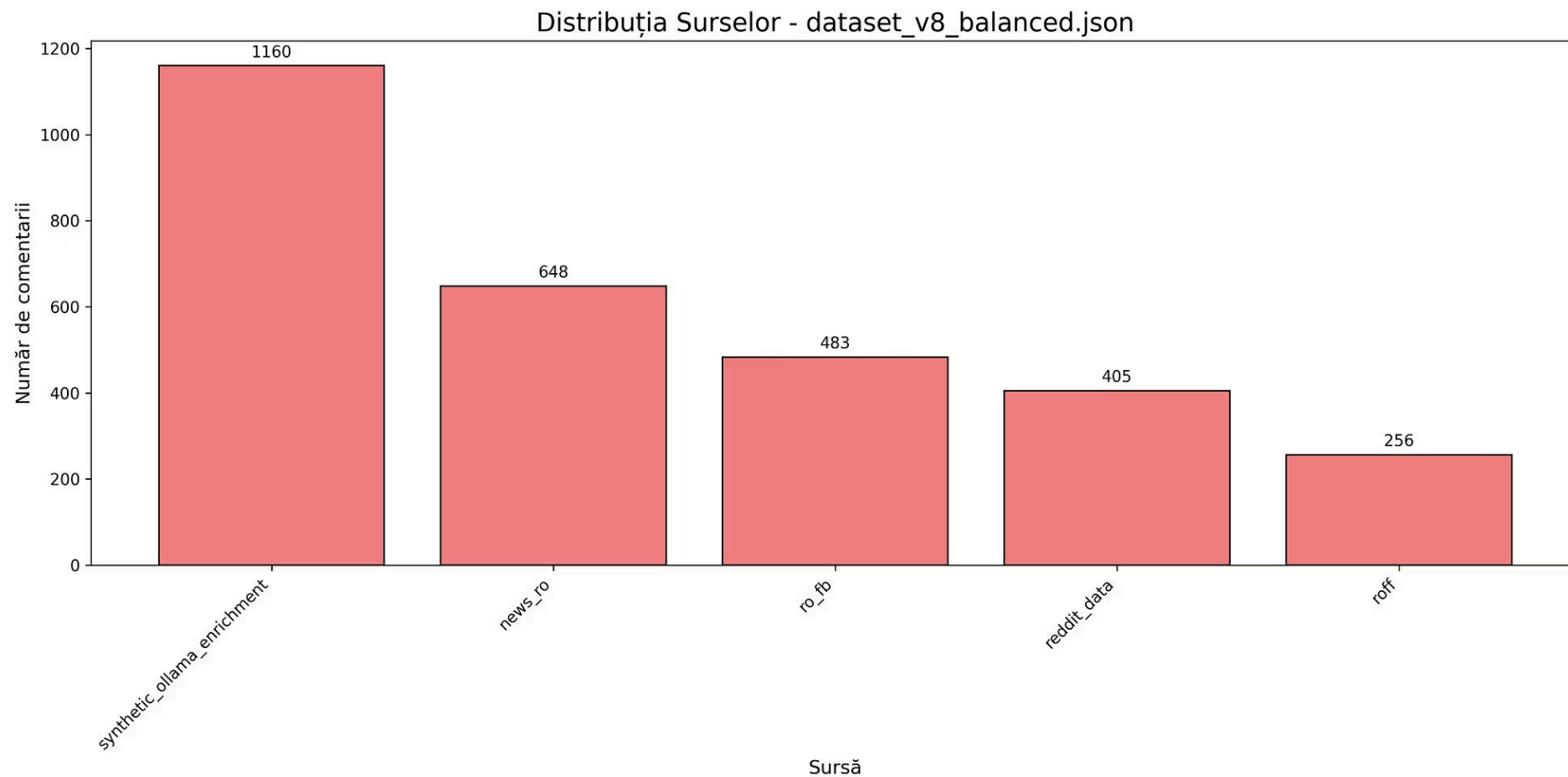
V8 Dataset Distribution



Reddit Dataset Sources



V8 Dataset Distribution



Most Frequent Words

Keyword	Reddit	News	Facebook	ROFF	V8 Balanced
dracu	631	24	145	395	174
prost	472	111	102	58	144
pula	455	68	25	274	146
muie	81	19	14	8	34
pizda	42	24	6	5	34
du-te	75	31	27	15	19

Conclusion & Future Work

Best Configuration

60% synthetic augmentation with XLM-RoBERTa achieves the strongest F1 (0.73).

Model Advantage

XLM-RoBERTa captures Romanian context far better than Linear SVM.

Limitations

Bias, label ambiguity, and synthetic data constraints remain challenges.

Future Directions

Multi-label classification, sarcasm detection, implicit hate, and REST API deployment.